III. Give short answer to the following questions.

28. What are the merits and demerits of one-way classification?
29. Write the model ANOVA table for one-way classification.
30. What are the values to be found for finding the test statistic in one-way classification?
31. What are the merits and demerits of two-way classification?
32. Write the model ANOVA table for two-way classification.
33. What are the components in two-way ANOVA?
34. What are the values to be found for finding the test statistic in two way classification?
35. Compare one-way and two-way ANOVA.

IV. Give detailed answer to the following questions.

36. In a sample of 8 observations, the sum of the squares of deviations of the sample values from its sample mean was 84.4. In another sample of 10 observations it was 102.6. Test whether the two population variances are equal at 5% level.
37. Two random samples gave the following results

Sample	Size	Sample mean	Sum of squares of deviations from the mean
I	10	15	90
II	12	14	108

Test whether the populations have same variances at 5% level of significance.

38. The following data refer to the yield of wheat in quintals on plots of equal area in two agricultural blocks A and B.

	Number of plots	Mean yield	Sample variance
Block A	8	60	50
Block B	6	51	40

Is the variance of yield for block A is greater than that of block B at 5% level of significance.

39. The calories contained in 1/2 cup servings of ice-creams selected randomly from two national brands are listed here. At 5% level of significance, is there sufficient evidence to conclude that the variance of calories is less for brand A than brand B?

Brand A	330	310	300	310	300	350	380	300	300
Brand B	300	300	270	290	310	370	300	310	250

40. The carbohydrates contained in servings of some randomly selected chocolate and non-chocolate candies are listed below. Is there sufficient evidence to conclude that the variance in carbohydrates varies between chocolate and non-chocolate candies? Use = 2%.

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